Report: Disaster Recovery & High Availability

1. Introduction

In any enterprise network, the availability and integrity of Active Directory (AD) are critical for identity, authentication, and resource management. Disaster Recovery (DR) planning and backup strategies must be meticulously implemented and periodically validated. This phase of the project focuses on creating a site-aware AD backup strategy tailored for domain controllers placed in distinct AD sites, and subsequently simulating a complete forest recovery in an isolated lab environment using the previously scheduled backups. This ensures organizational readiness for catastrophic AD failure scenarios and validates the reliability of both the backup and recovery process.

2. Objective

The primary objective of Task 54 was to establish a site-aware AD backup strategy wherein each site performs independent and scheduled backups, ensuring redundancy. For Task 55, the goal was to simulate a complete forest-wide disaster by manually deleting AD-related NTDS files, and then recover the domain by restoring from a system state backup in an isolated environment. The overall intent was to validate that AD services could be recovered fully, accurately, and within an acceptable timeframe, thereby demonstrating a proven disaster recovery capability.

3. Tools used

- **Windows Server 2022 (Domain Controller)**
- **Windows 10 (Domain-joined Client)**
- **PowerShell**
- **wbadmin (Windows Backup Tool)**
- **Task Scheduler**
- **Hyper-V/VMware (Lab Environment)**

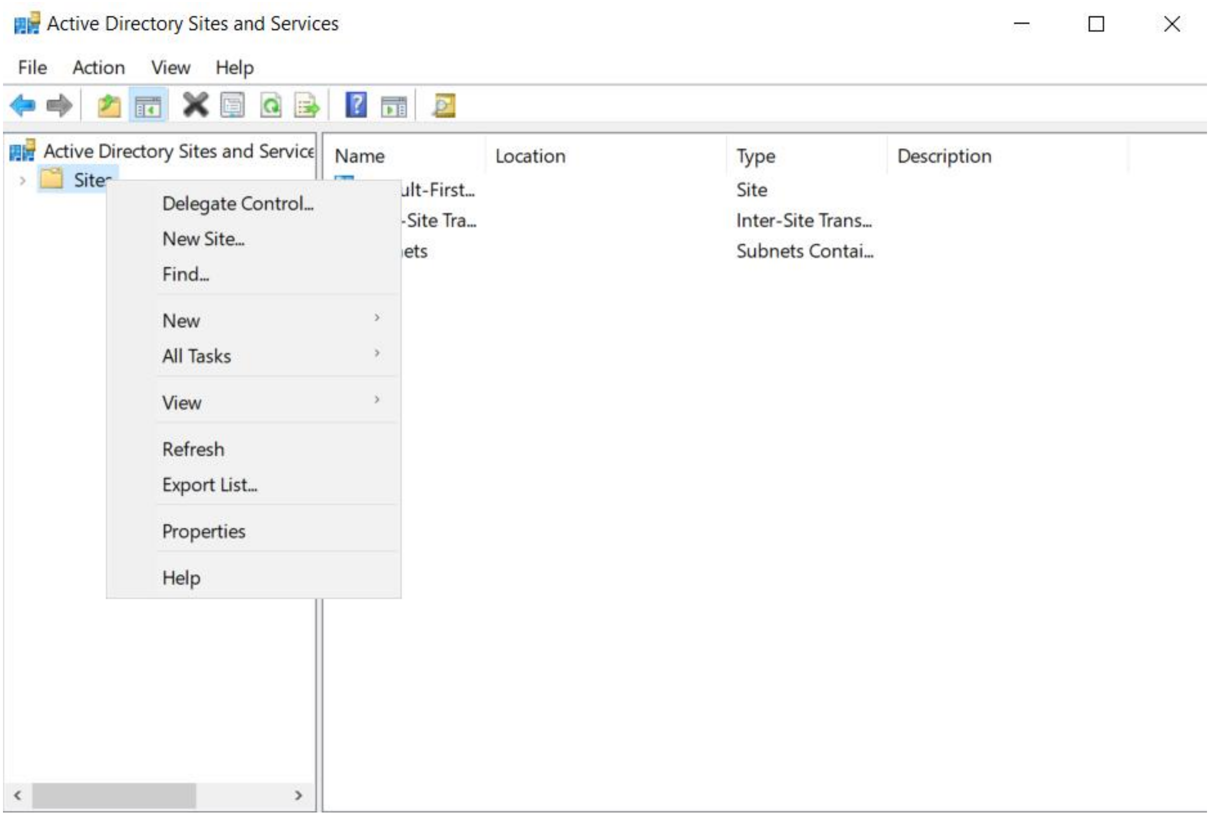
- **NTDS directory management commands**

4. Methodology

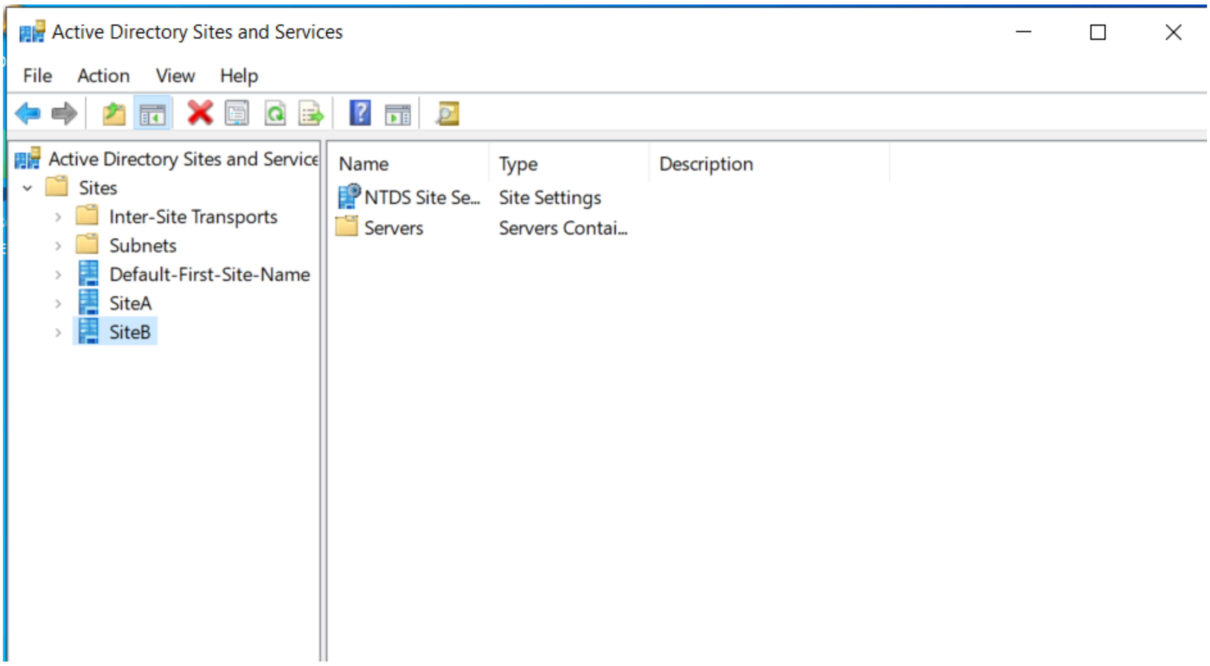
4.1 Implementing a site-aware AD backup strategy

Two AD Sites were defined within Active Directory Sites and Services: SiteA and SiteB. The existing domain controller was moved under SiteA. To support independent backup configurations, two separate directories (C:\SiteABackups and C:\SiteBBackups) were created. PowerShell scripts were written for both sites using the wbadmin start systemstatebackup command, directing backups to an external disk volume (E:). These scripts (SiteA_Backup.ps1 and SiteB_Backup.ps1) were saved under C:\Scripts\.

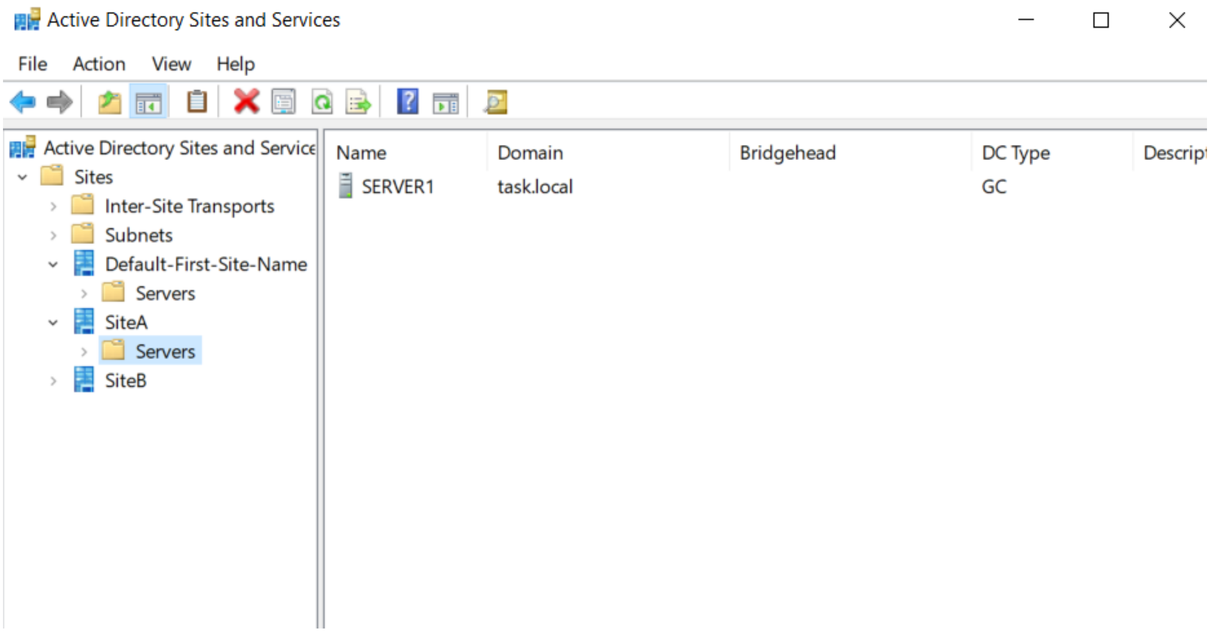
Task Scheduler was then used to automate the backup process. For SiteA, a scheduled task was configured to run daily at 2:00 AM, with the backup script executed using elevated privileges regardless of user logon state. The task used powershell.exe -ExecutionPolicy Bypass -File "C:\Scripts\SiteA_Backup.ps1" as its action. All required conditions for idle time, power state, and on-demand execution were configured. A test run of the task confirmed the creation of a successful system state backup on the designated external volume.



(Create Two Sites in AD)



(Created 2 new sites)



(Move your server to SiteA)

```
PS C:\Users\Administrator> New-Item -ItemType Directory -Path "E:\SiteA"


    Directory: E:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         5/16/2025  11:28 AM                SiteA


PS C:\Users\Administrator> New-Item -ItemType Directory -Path "E:\SiteB"


    Directory: E:\


Mode                 LastWriteTime         Length Name
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d-----         5/16/2025  11:28 AM                SiteB
```
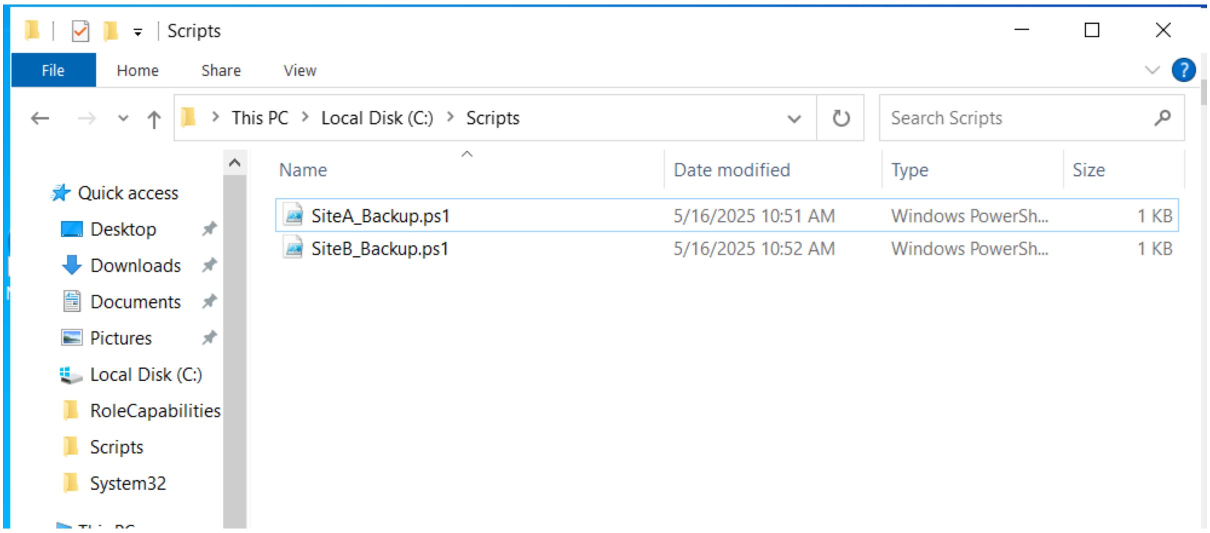
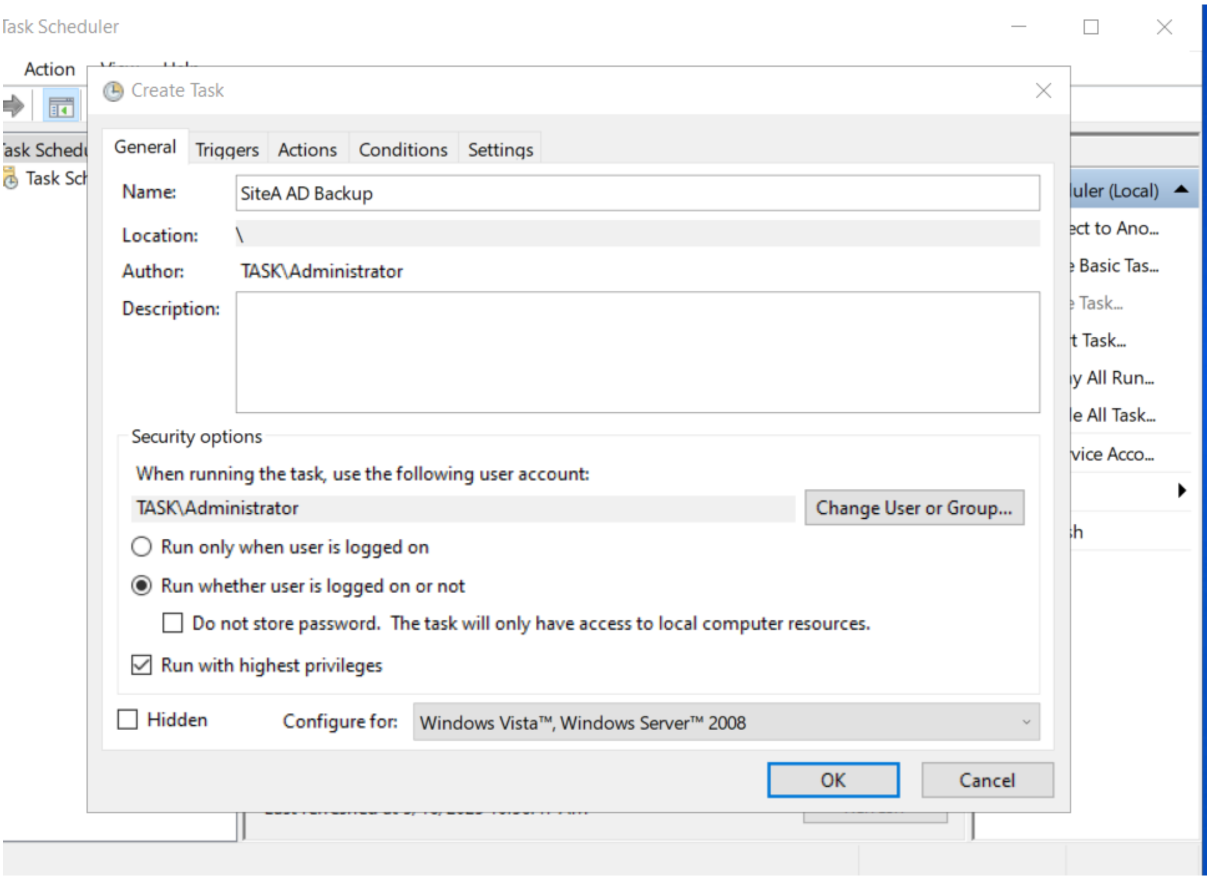
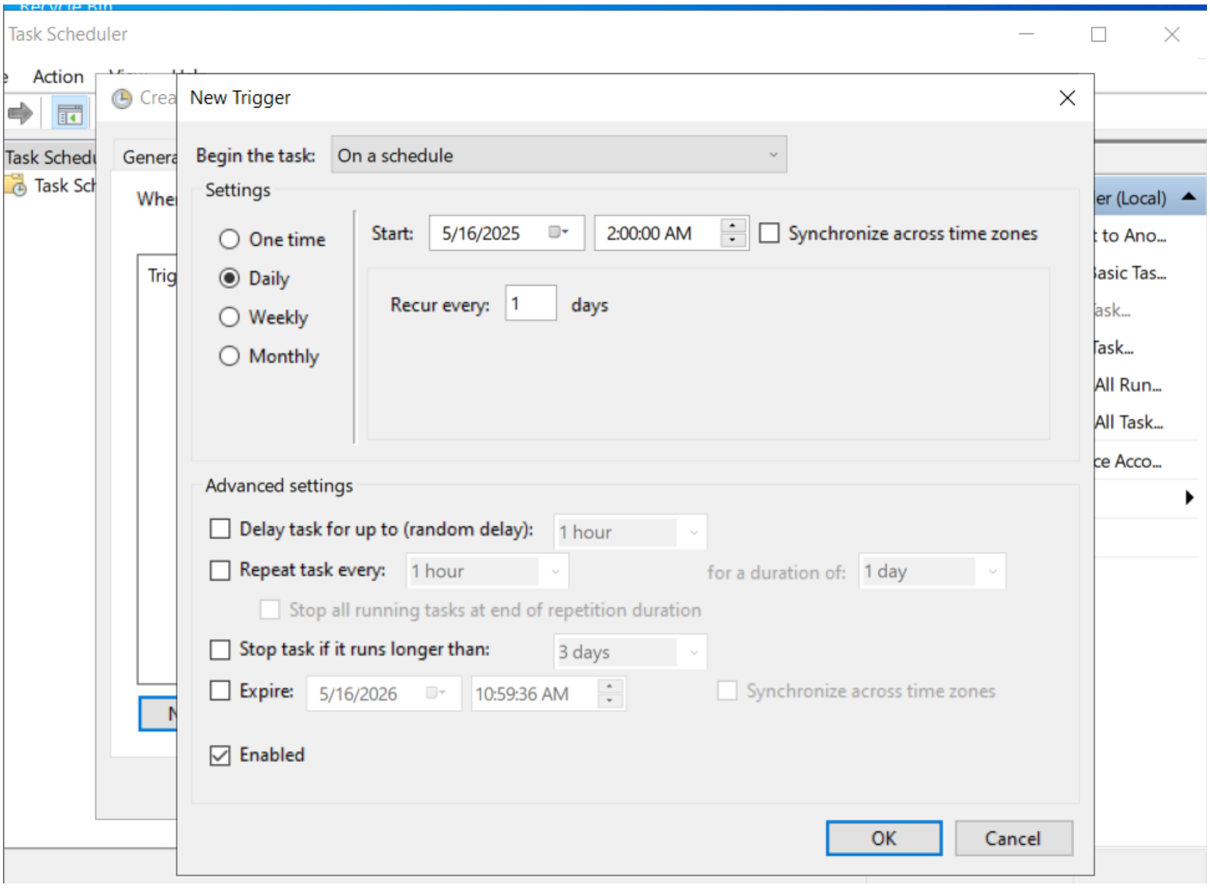

(Create Backup Folders)



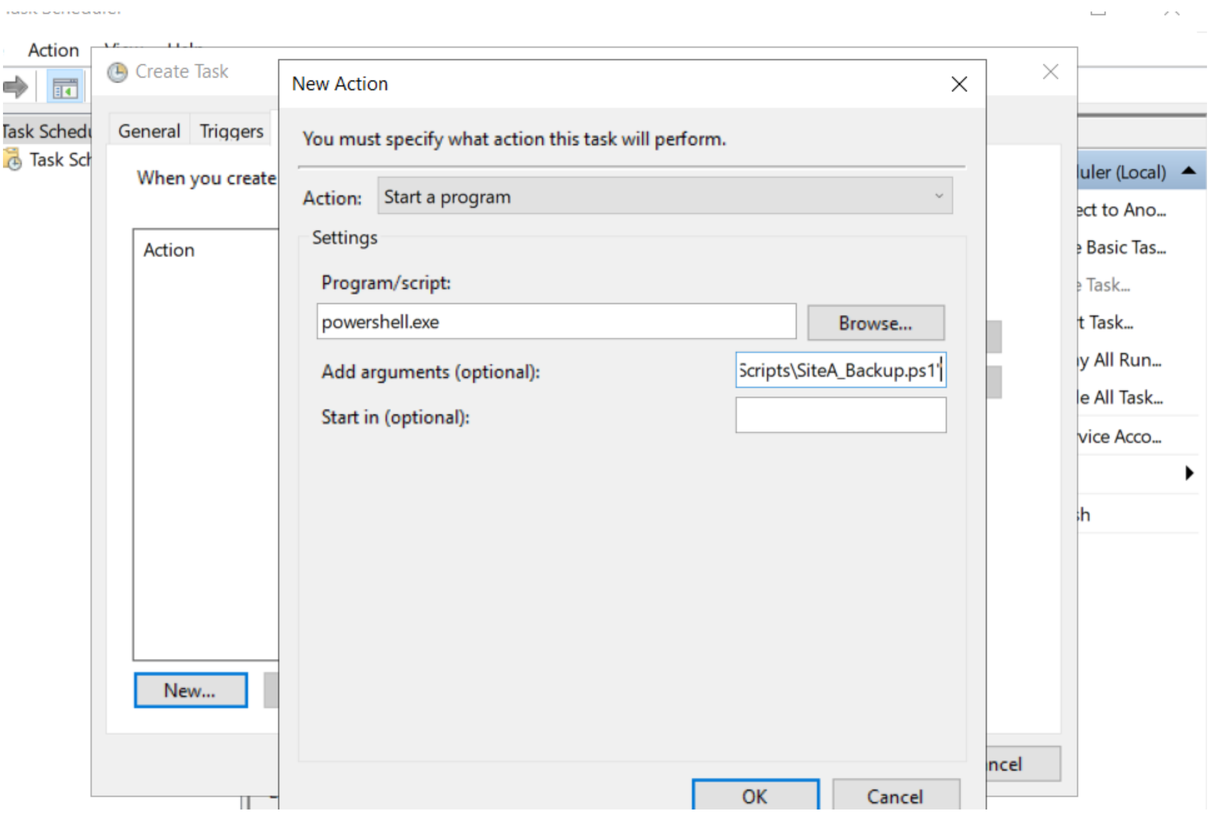
(Create PowerShell Backup Scripts)



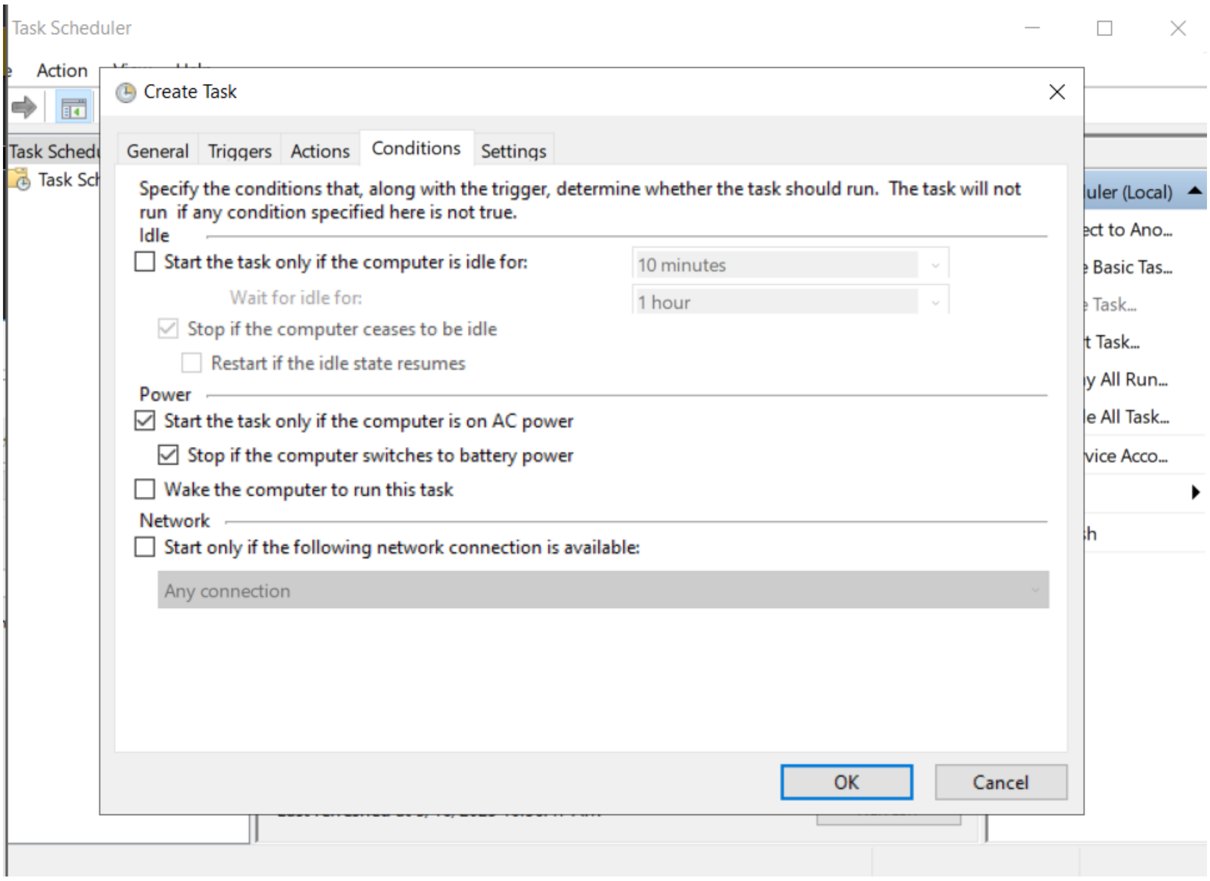
(Create task)



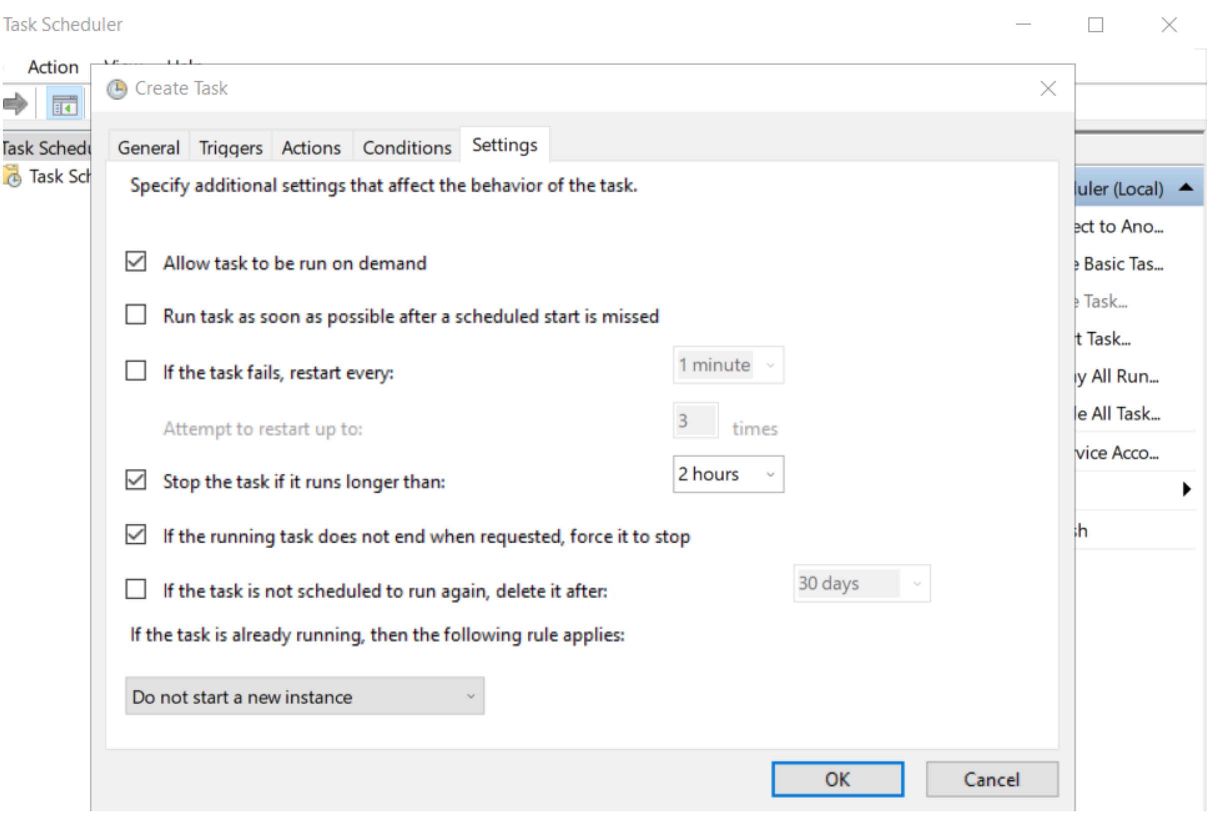
(Set trigger)



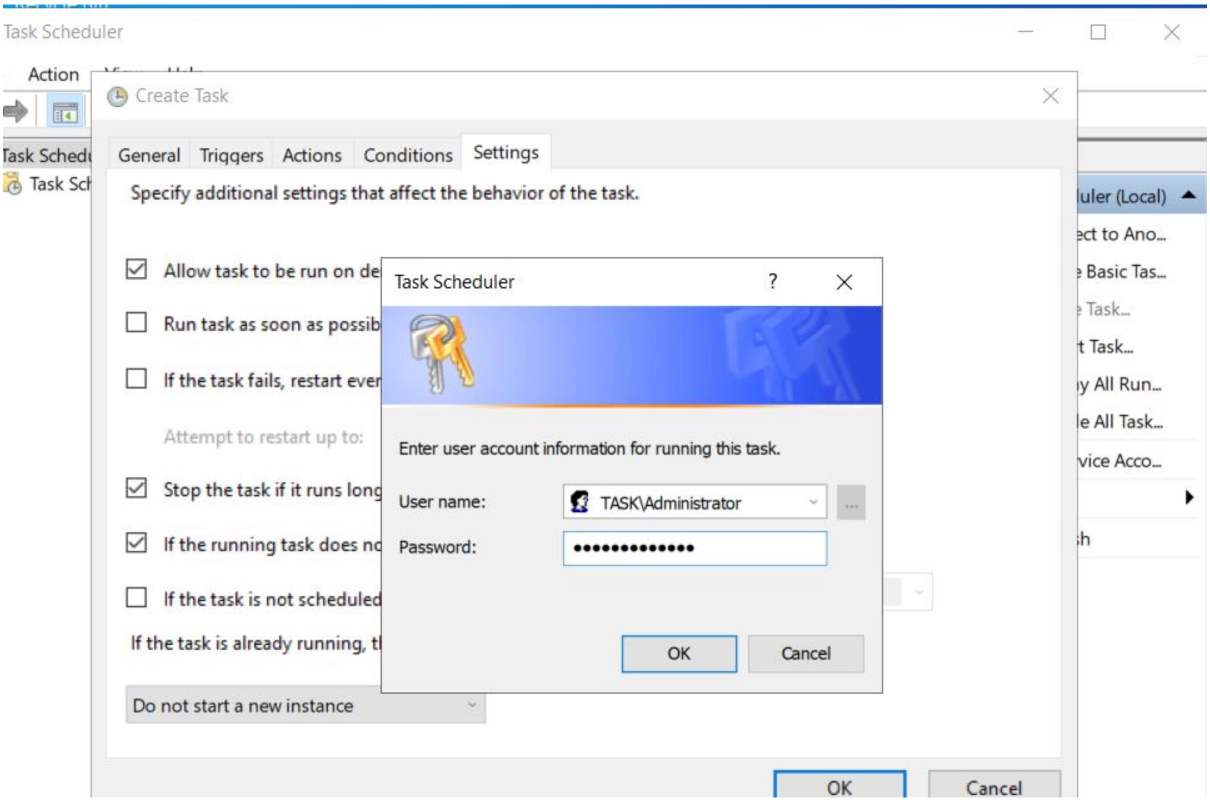
(Action to perform on trigger)



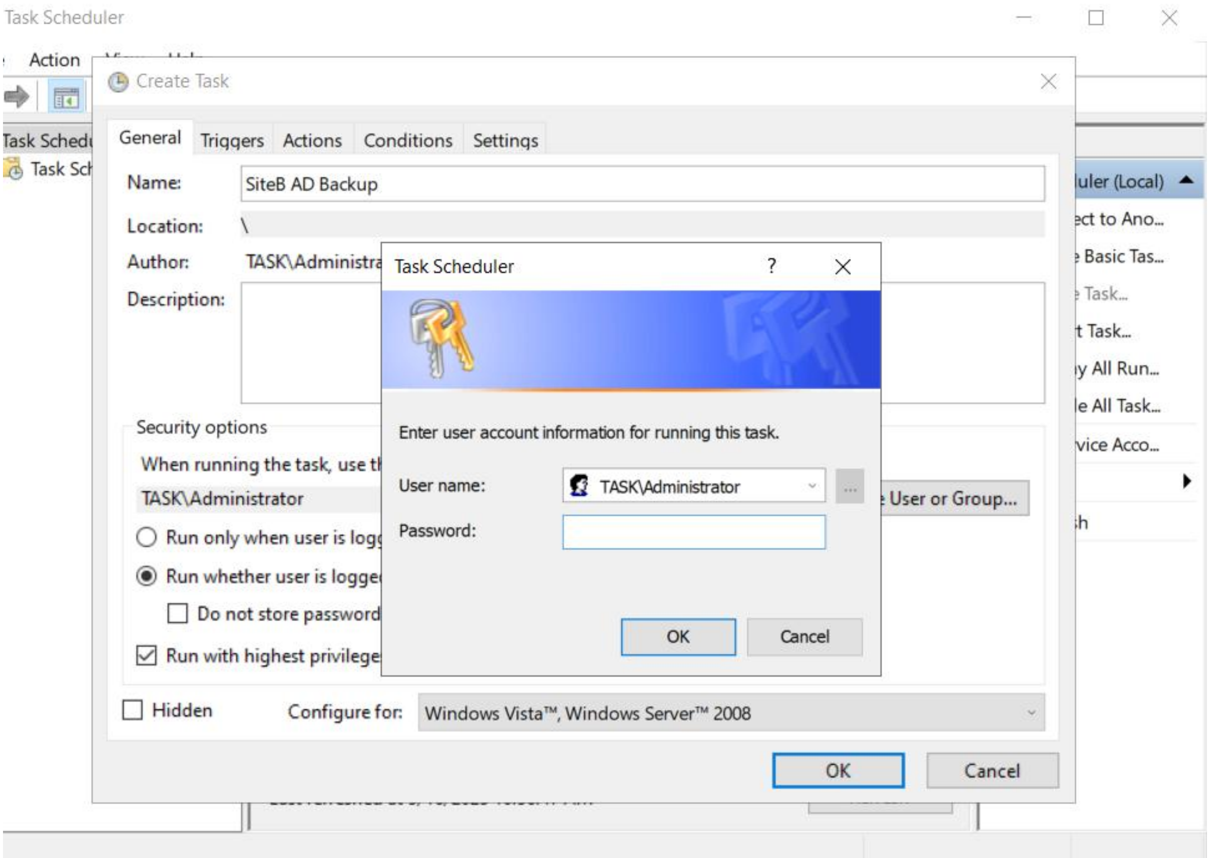
(Set conditions)



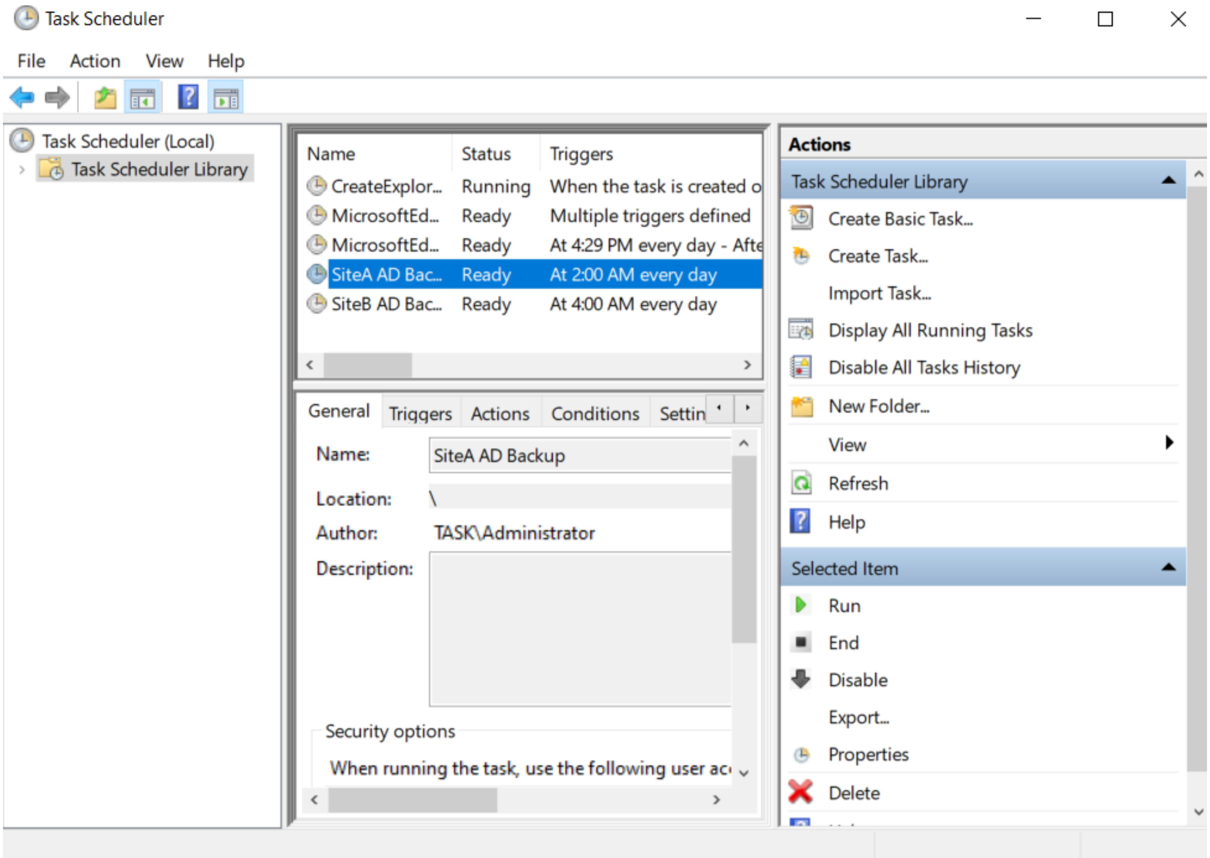
(Final configuration)



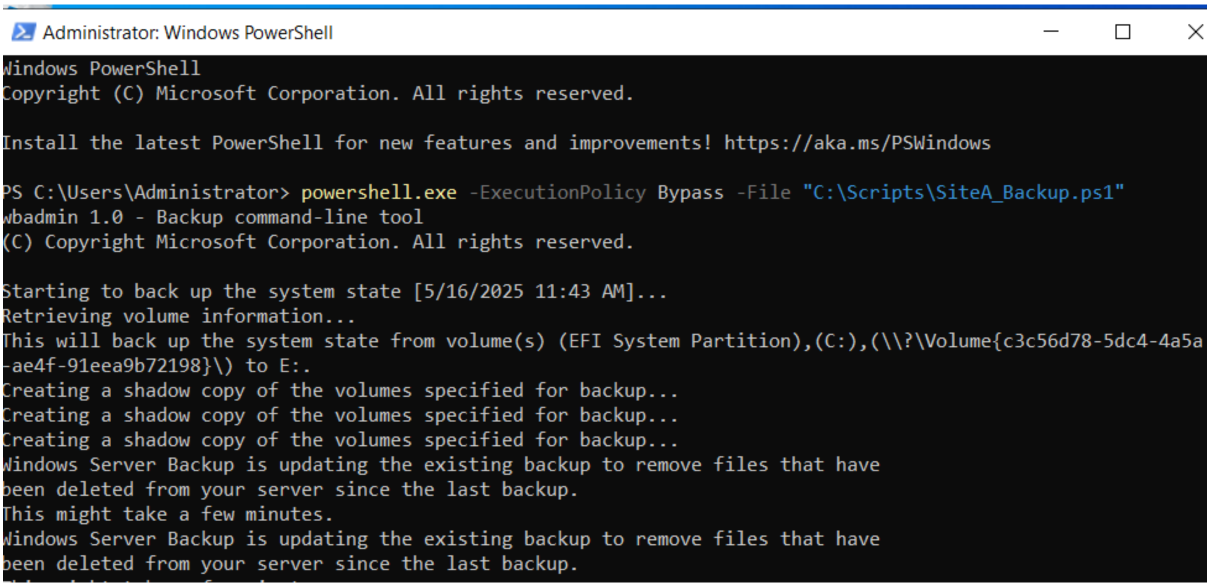
(Schedule the task)



(Same steps were repeated for siteB)



(Successfully scheduled both tasks)

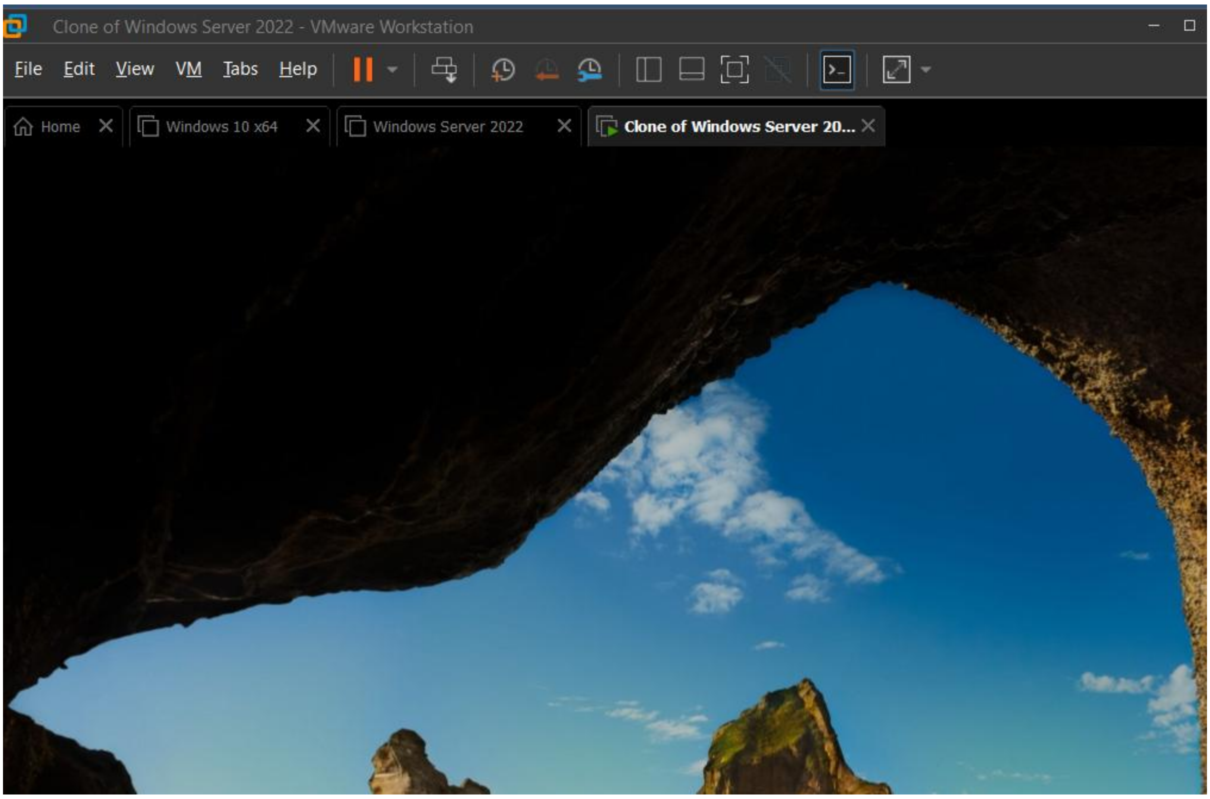


(Trigger manually for testing and it is working properly)

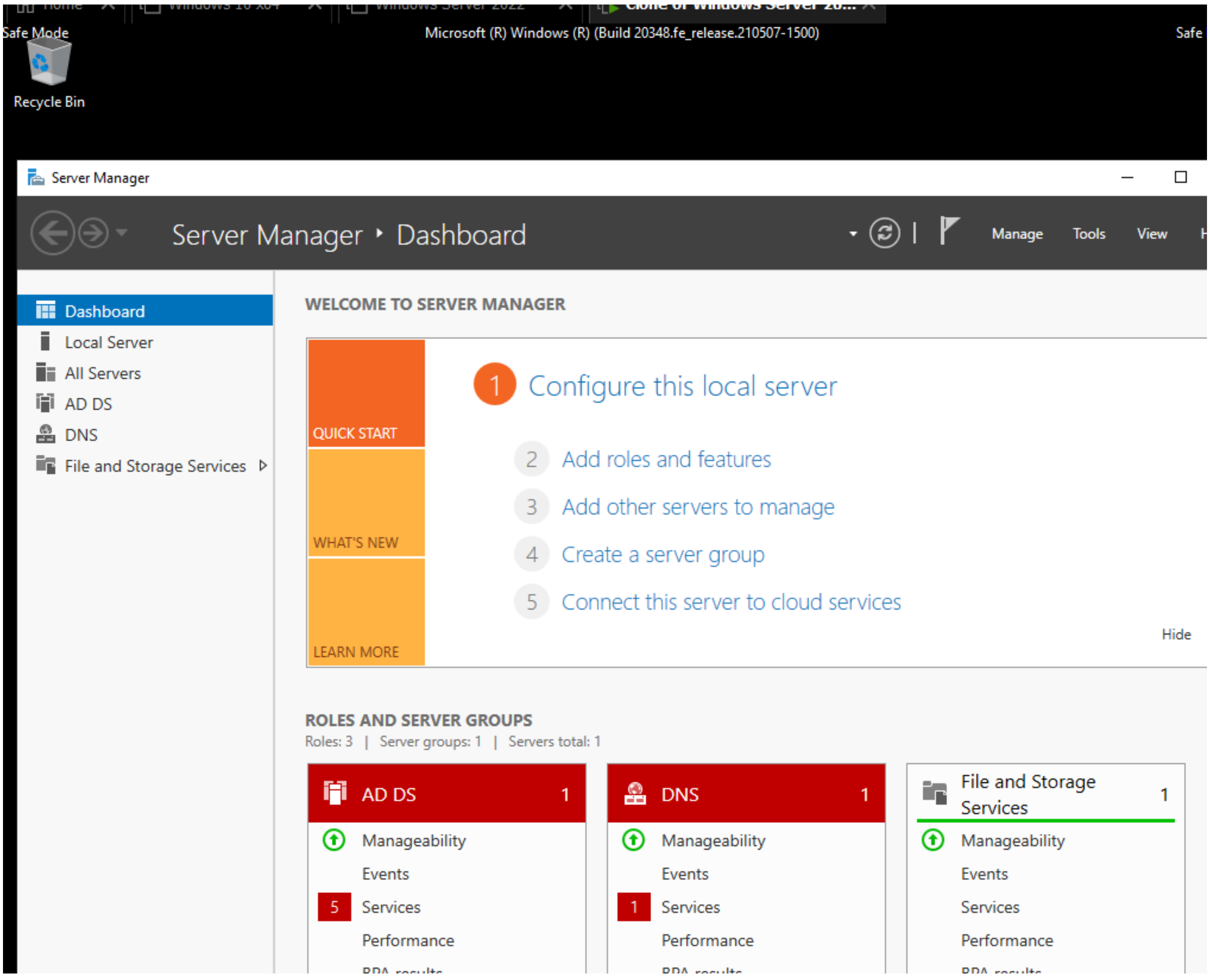
4.2 Testing a full AD forest recovery in an isolated lab

The system was rebooted into Directory Services Repair Mode (DSRM) to simulate a catastrophic failure. Using takeown and icacls, full administrative permissions were granted on C:\Windows\NTDS. The NTDS folder contents were then deleted via PowerShell to fully remove Active Directory data. A normal reboot confirmed that AD services were broken, verifying that the simulation of corruption was effective.

Recovery was initiated by identifying the correct backup version using wbadmin get versions, followed by executing wbadmin start systemstaterecovery with the appropriate version ID. After rebooting, the system completed the restore process and brought the Domain Controller back to its pre-disaster state. Functionality of AD, DNS, and SYSVOL replication were verified post-recovery. The test validated that the backup strategy was not only operational but also reliable under real-world failure conditions.



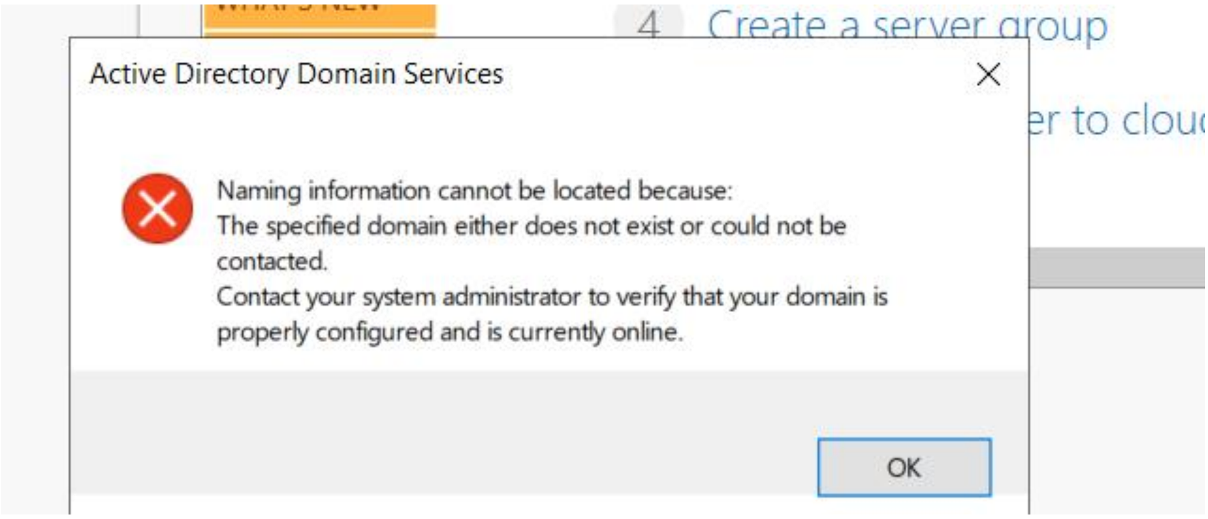
(Create a full clone of server for testing recovery)



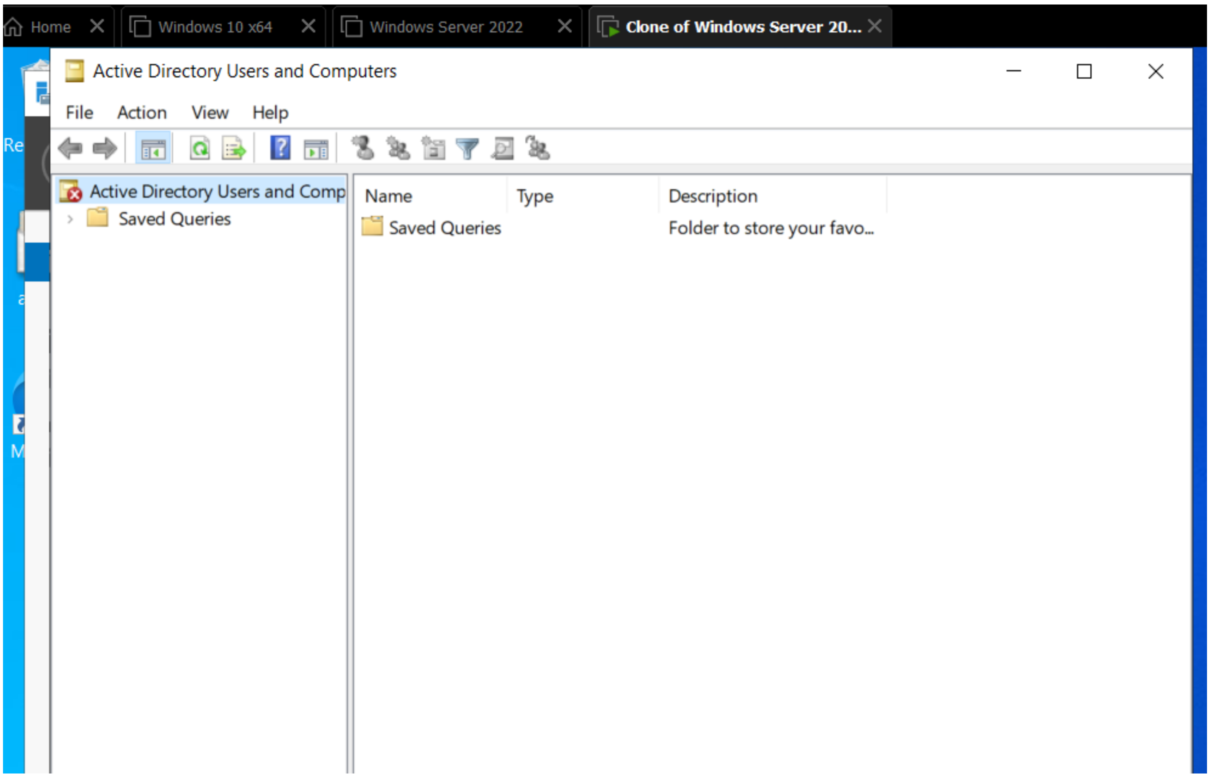
(Open cloned server in recovery mode)

```
PS C:\Users\Administrator.SERVER1.001> takeown /f C:\Windows\NTDS /r /d y

SUCCESS: The file (or folder): "C:\Windows\NTDS" now owned by user "SERVER1\Administrator".

SUCCESS: The file (or folder): "C:\Windows\NTDS\edb.chk" now owned by user "SERVER1\Administrator".

SUCCESS: The file (or folder): "C:\Windows\NTDS\edb.log" now owned by user "SERVER1\Administrator".

SUCCESS: The file (or folder): "C:\Windows\NTDS\edbres00001.jrs" now owned by user "SERVER1\Administrator".

SUCCESS: The file (or folder): "C:\Windows\NTDS\edbres00002.jrs" now owned by user "SERVER1\Administrator".

SUCCESS: The file (or folder): "C:\Windows\NTDS\edbtmp.log" now owned by user "SERVER1\Administrator".

SUCCESS: The file (or folder): "C:\Windows\NTDS\ntds.dit" now owned by user "SERVER1\Administrator".

SUCCESS: The file (or folder): "C:\Windows\NTDS\ntds.jfm" now owned by user "SERVER1\Administrator".
PS C:\Users\Administrator.SERVER1.001>
PS C:\Users\Administrator.SERVER1.001> icacls C:\Windows\NTDS /grant administrators:F /t
processed file: C:\Windows\NTDS
processed file: C:\Windows\NTDS\edb.chk
processed file: C:\Windows\NTDS\edb.log
processed file: C:\Windows\NTDS\edbres00001.jrs
processed file: C:\Windows\NTDS\edbres00002.jrs
processed file: C:\Windows\NTDS\edbtmp.log
processed file: C:\Windows\NTDS\ntds.dit
processed file: C:\Windows\NTDS\ntds.jfm
Successfully processed 8 files; Failed processing 0 files
```

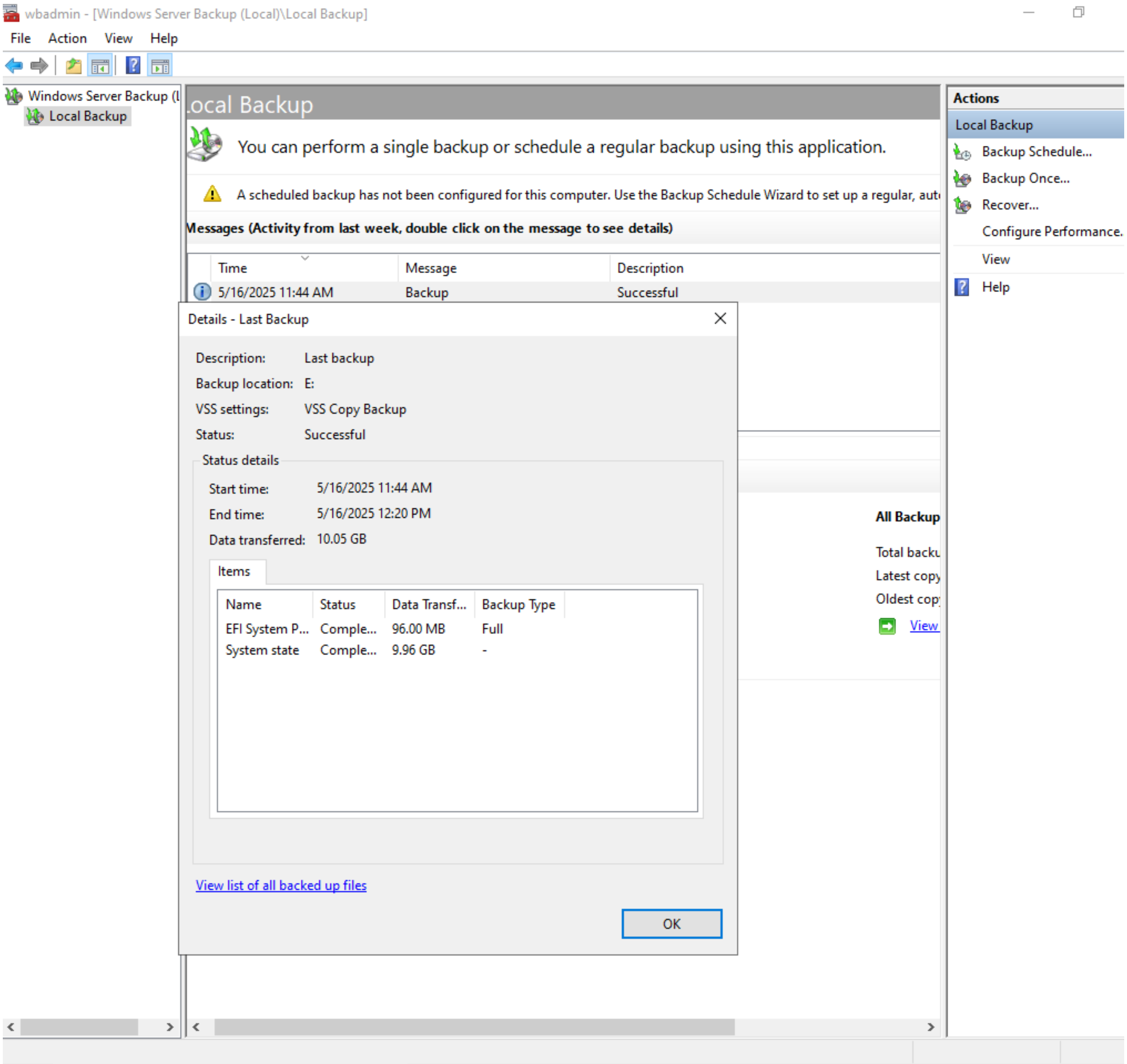
(deleted the **Active Directory database files** (e.g., ntds.dit, logs))



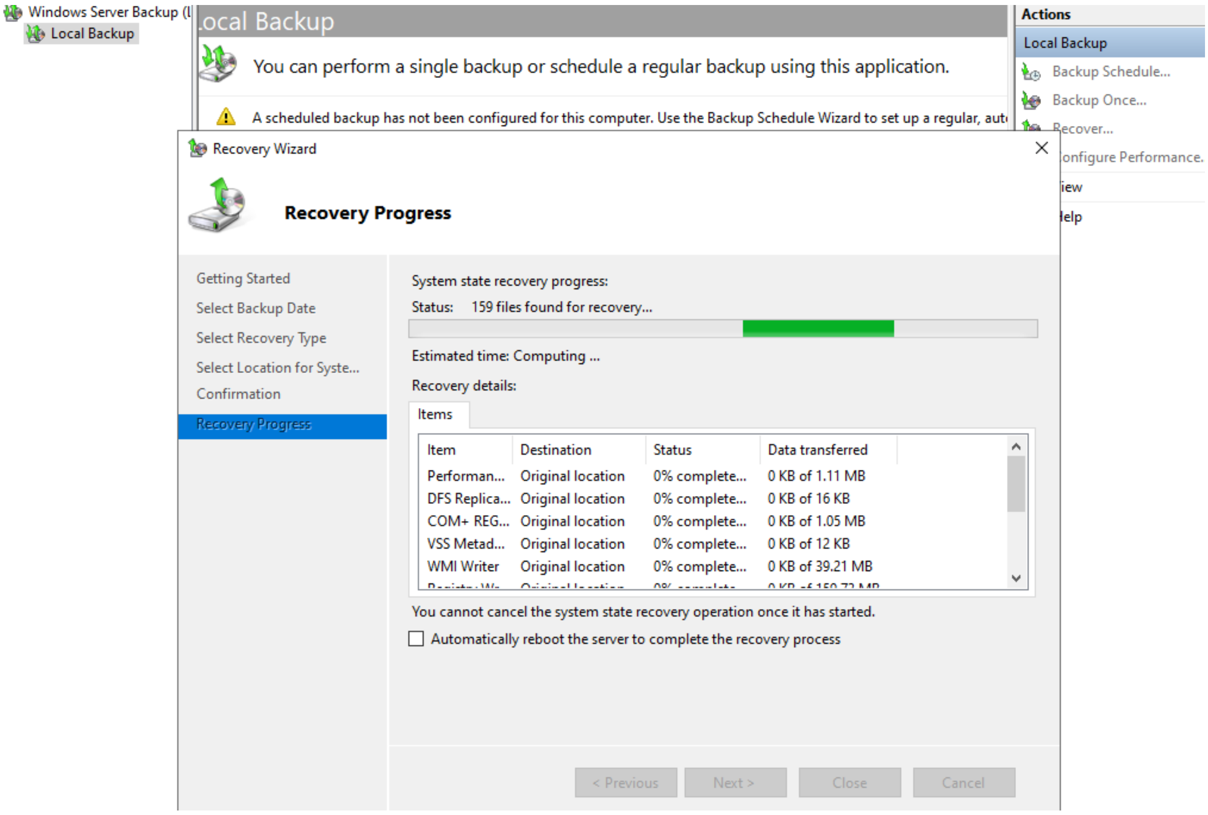
(Ad is damaged)



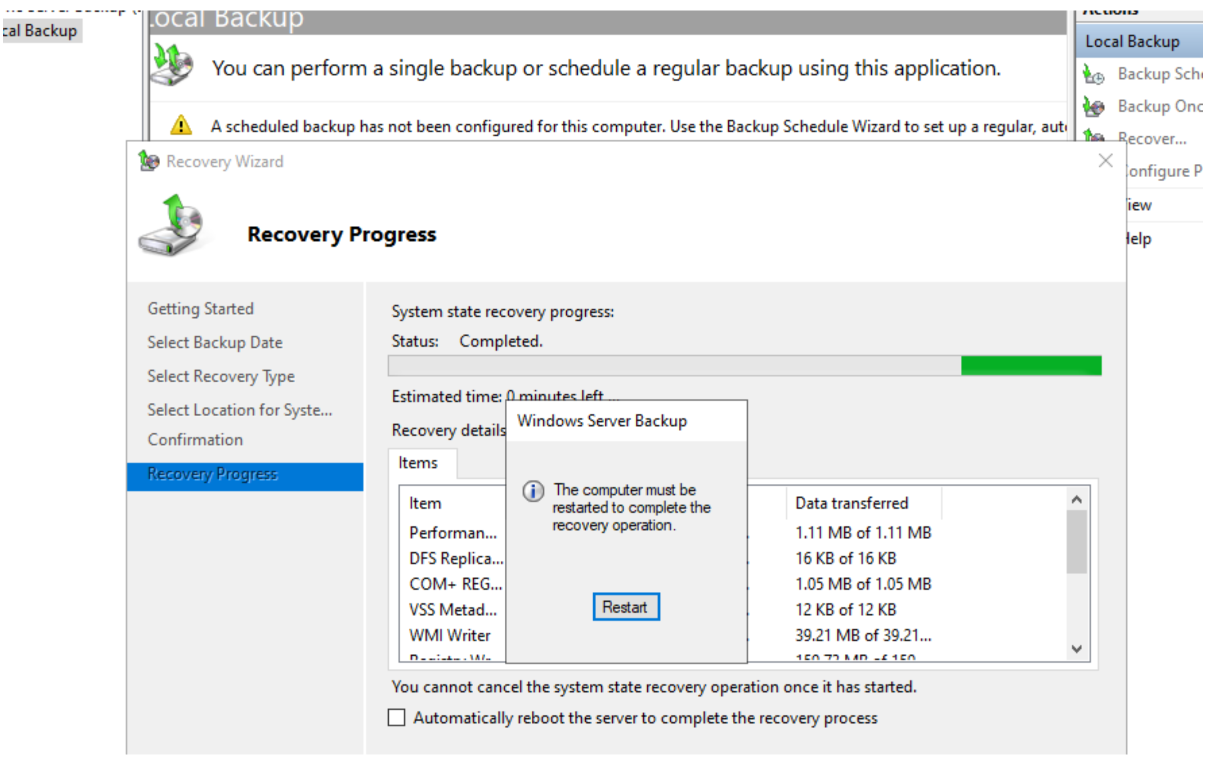
(Its empty)



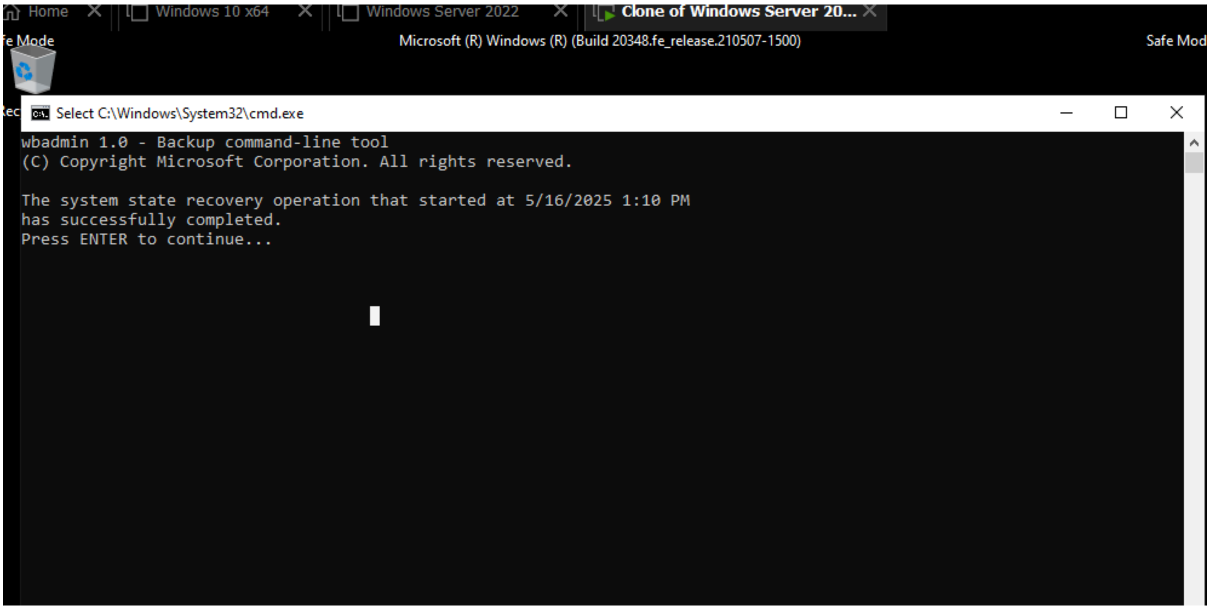
(Now let’s recover it with the backup which was taken in previous task)



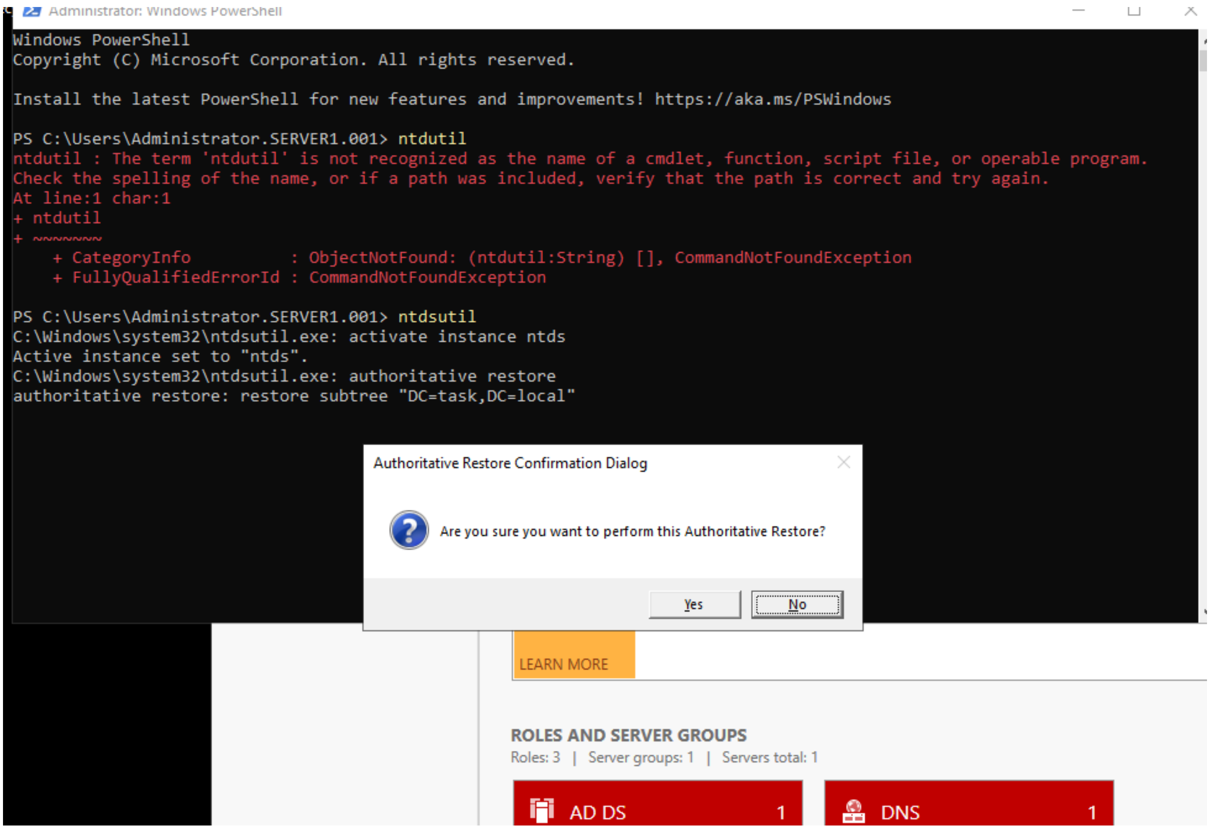
(Backup started)



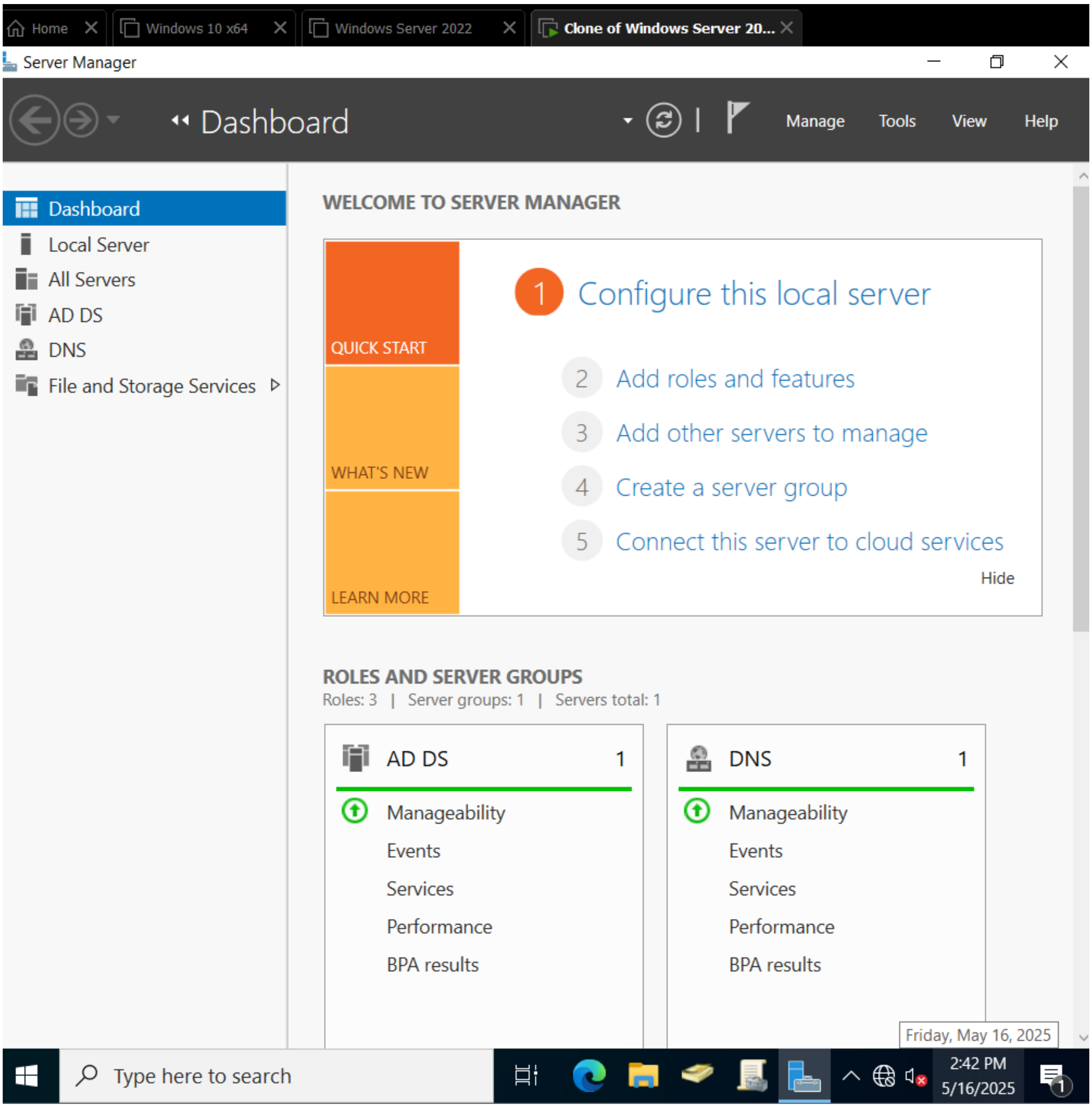
(Backup completed)



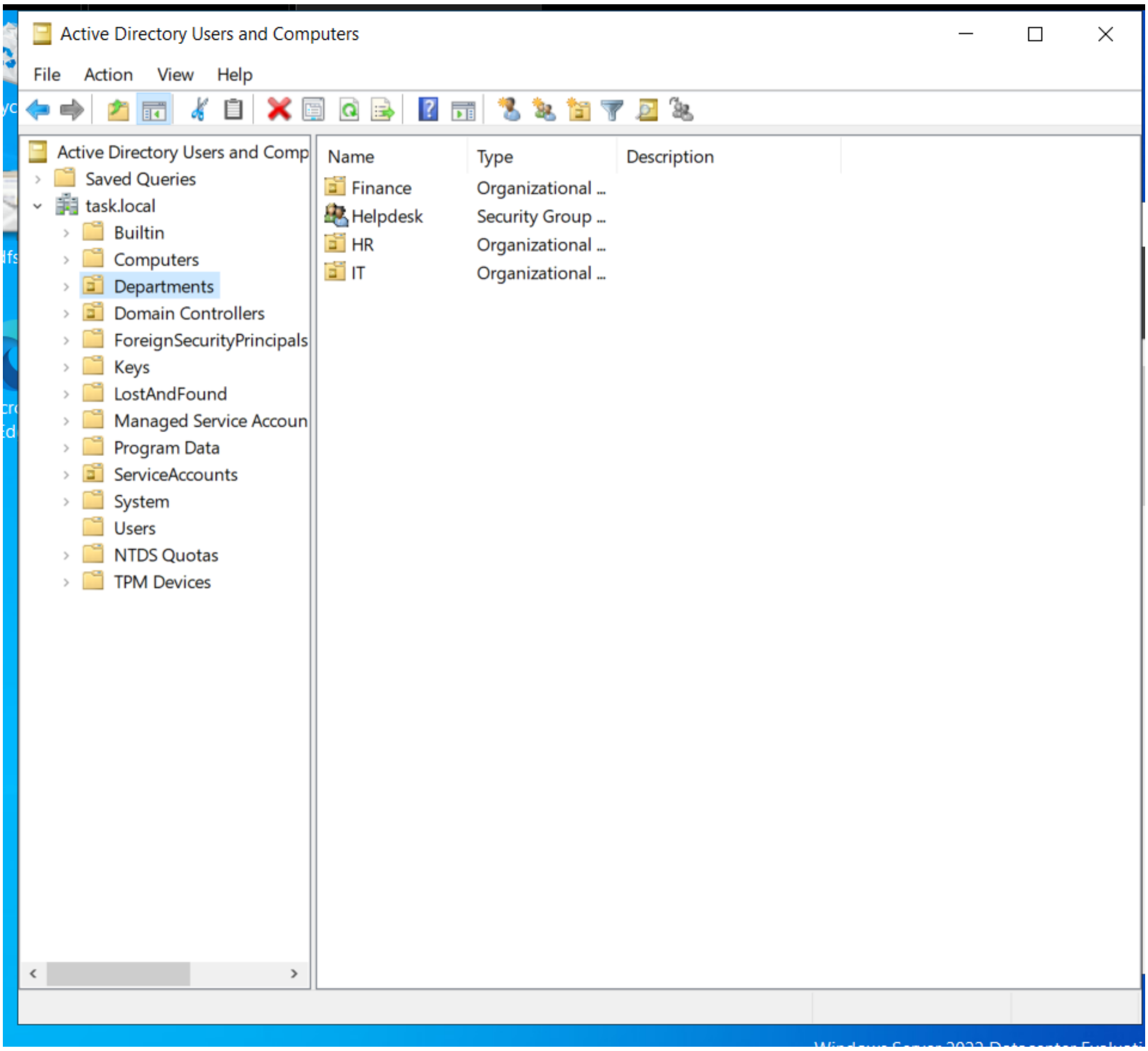
(Done successfully)



(authoritative restore is used)



(AD is recovered)



(Back to the original state)

5. Recommendations

- Always store system state backups on a separate volume or external disk to comply with wbadmin constraints.
- Schedule staggered backups across sites to avoid simultaneous I/O load and allow flexibility in recovery timelines.
- Consider automating post-backup integrity checks using PowerShell logging.
- Ensure all Domain Controllers are members of the "Backup Operators" group or have delegated permissions.

- Regularly test forest recovery scenarios in an isolated environment using cloned or snapshot VMs.
- Maintain documentation of all Domain Controller backup locations and scheduled job configurations.

6. Conclusion

The execution of Task 54 successfully established a site-aware backup strategy using independent schedules and scripts, simulating a geographically distributed backup approach. This ensures redundancy and quicker recovery in the event of regional outages. Task 55 further validated the resilience of the environment by simulating complete Active Directory corruption and demonstrating a full forest recovery using the previously configured backup. The steps taken, from DSRM boot, NTDS deletion, and system state restore, ensured that AD was restored exactly to its pre-disaster configuration. This end-to-end lab proves that the disaster recovery plan is not only viable but also executable within a contained lab setup, giving confidence in AD recovery preparedness.